DATA STRUCTURES LAB RECORD

Implementation of Hashing Technique (Linear Probing) in C

Source Code (HASHING.C)

```
#include <stdio.h>
#include <stdlib.h>
#define TABLE_SIZE 10

int h[TABLE_SIZE] = {NULL};

void insert() {
    int key, index, i, flag = 0, hkey;
    printf("\nenter a value to insert into hash table\n");
    scanf("%d", &key);
    hkey = key % TABLE_SIZE;
    for (i = 0; i < TABLE_SIZE; i++) {
        index = (hkey + i) % TABLE_SIZE;
        if (h[index] == NULL) {
            h[index] = key;
            break;
        }
    }
    if (i == TABLE_SIZE)
        printf("\nelement cannot be inserted\n");
}

void search() {
    int key, index, i, flag = 0, hkey;
    printf("\nenter search element\n");
    scanf("%d", &key);
    hkey = key % TABLE_SIZE;
    for (i = 0; i < TABLE_SIZE; i++) {
        index = (hkey + i) % TABLE_SIZE;
        if (h[index] == key) {
            printf("value is found at index %d", index);
            break;
        }
    }
    if (i == TABLE_SIZE)
        printf("\n value is not found\n");
}

void display() {
    int i;
    printf("\nelements in the hash table are \n");
    for (i = 0; i < TABLE_SIZE; i++)
        printf("\nat index %d \t value = %d", i, h[i]);
}

int main() {
    int opt, i;
    while (1) {
        printf("\nPress 1. Insert\t 2. Display \t3. Search \t4.Exit \n");
        scanf("%d", &opt);
        switch (opt) {
            case 1:
                insert();
                break;
            case 2:
                display();
                break;
            case 3:
                search();
                break;
            case 4:
                exit(0);
        }
    }
}
```

Sample Output

```
Press 1. Insert 2. Display 3. Search 4.Exit
1
enter a value to insert into hash table
34

Press 1. Insert 2. Display 3. Search 4.Exit
1
enter a value to insert into hash table
45

Press 1. Insert 2. Display 3. Search 4.Exit
2
elements in the hash table are
at index 0 value = 0
at index 1 value = 0
at index 2 value = 0
at index 3 value = 0
at index 4 value = 34
at index 5 value = 45
at index 6 value = 0
at index 7 value = 0
at index 8 value = 0
at index 9 value = 0

Press 1. Insert 2. Display 3. Search 4.Exit
3
enter search element
45
value is found at index 5

Press 1. Insert 2. Display 3. Search 4.Exit
4
```